

1) The assembly language source code

```
6  .comm buf, 4096, 4
7  .text
8  .globl _start
9  _start:
10  loop:
11      # read(0, buf, 4096)
12      movq $0, %rax
13      movq $0, %rdi
14      movq $buf, %rsi
15      movq $4096, %rdx
16      syscall
17
18      # if n <= 0, exit
19      cmpq $0, %rax
20      jle exit
21      movq %rax, %r12
22
23      # write(1, buf, n)
24      movq $1, %rax
25      movq $1, %rdi
26      movq $buf, %rsi
27      movq %r12, %rdx
28      syscall
29      jmp loop
30
31  exit:
32      # _exit(n)
33      movq %rax, %rdi
34      movq $60, %rax
35      syscall
36
```

2) Build process

```
root@MSI:/mnt/c/Github/OperatingSystem/pset7# as --64 -o catasm.o catasm.S
root@MSI:/mnt/c/Github/OperatingSystem/pset7# ld -m elf_x86_64 -o catasm catasm.o
root@MSI:/mnt/c/Github/OperatingSystem/pset7# ./catasm
hi testing
hi testing
oh good
oh good
it's working
it's working
```

- 3) A screenshot of running the cat program, and echo \$?

```
root@MSI:/mnt/c/Github/OperatingSystem/pset7# ./catasm
hi
hi
^Z
[3]+  Stopped                  ./catasm
root@MSI:/mnt/c/Github/OperatingSystem/pset7# echo $?
148
root@MSI:/mnt/c/Github/OperatingSystem/pset7# echo "hi" | ./catasm
hi
root@MSI:/mnt/c/Github/OperatingSystem/pset7# echo $?
0
```

- 4) Under strace

```

root@MSI:/mnt/c/Github/OperatingSystem/pset7# yes "Testing Testing Testing Testing." | head -n 500 > bigfile.txt
root@MSI:/mnt/c/Github/OperatingSystem/pset7# ls -l
total 40
-rwxrwxrwx 1 root root 20500 Dec 20 12:02 bigfile.txt
-rwxrwxrwx 1 root root 4864 Dec 20 11:50 catasm
-rwxrwxrwx 1 root root 644 Dec 20 11:31 catasm.S
-rwxrwxrwx 1 root root 904 Dec 20 11:50 catasm.o
-rwxrwxrwx 1 root root 0 Dec 20 10:59 output.txt
root@MSI:/mnt/c/Github/OperatingSystem/pset7# strace ./catasm < bigfile.txt > /dev/null
execve("./catasm", [ "./catasm" ], 0x7ffcf2e893e0 /* 31 vars */) = 0
read(0, "Testing Testing Testing Testing "..., 4096) = 4096
write(1, "Testing Testing Testing Testing "..., 4096) = 4096
read(0, "ng.\nTesting Testing Testing Test"..., 4096) = 4096
write(1, "ng.\nTesting Testing Testing Test"..., 4096) = 4096
read(0, "esting.\nTesting Testing Testing "..., 4096) = 4096
write(1, "esting.\nTesting Testing Testing "..., 4096) = 4096
read(0, "ng Testing.\nTesting Testing Test"..., 4096) = 4096
write(1, "ng Testing.\nTesting Testing Test"..., 4096) = 4096
read(0, "esting Testing.\nTesting Testing "..., 4096) = 4096
write(1, "esting Testing.\nTesting Testing "..., 4096) = 4096
read(0, "ng Testing Testing.\n", 4096) = 20
write(1, "ng Testing Testing.\n", 20) = 20
read(0, "", 4096) = 0
exit(0) = ?
+++ exited with 0 +++
root@MSI:/mnt/c/Github/OperatingSystem/pset7#

```

[illegible]